

AXIS

ERAP1 × Axial Spondyloarthritis

Independent Scientific Review Pack

Version 1.0 | 6 October 2026 | Independent review pending

Evidence cutoff: **2026-10-05**

Frozen scientific/runtime state:

d11128538a2723737eced46be73d40cef56f43db

AXIS is a scientific decision system designed to make the reasoning between evidence, uncertainty and the next discriminating experiment explicit and auditable. In this case, it links source-scoped ERAP1 evidence to categorical assessments, competing explanations and conditional experimental consequences. It preserves distinctions between source assertions, calculations, inferences and proposed experiments. The present pack summarizes one frozen exploratory decision for expert criticism; it does not present AXIS as an autonomous scientist or treat automated reproducibility as scientific approval.

Review purpose

This is a request for independent scientific criticism, not endorsement.

We are not asking whether you endorse AXIS. We are asking where this scientific decision may be wrong.

Please seek factual errors, overclaims, material missing evidence within scope, inappropriate evidence transfer, flawed assay comparisons, mechanistic reasoning errors, incorrect uncertainty prioritization and weaknesses in the next experiment.

Reading route

Read pages 2-5 for the decision in approximately 10 minutes. Use the questions, primary references and immutable repository links on pages 6-7 for a 30-60 minute audit.

Status and boundary

AXIS has not been independently scientifically validated.

This document summarizes a frozen scientific decision analysis for independent review. It is not clinical guidance, evidence of therapeutic efficacy, regulatory advice, or an endorsement of ERAP1 as a validated therapeutic target.

Decision snapshot

Genetics → Mechanism → Structure → Chemistry → Cellular pharmacology → Target engagement → Disease translation

Layer	Frozen assessment / scope
Genetics	ERAP1 × HLA association recorded; association does not establish therapeutic direction.
Mechanism	Peptide-processing context; functional edge: INSUFFICIENT.
Structure	13 mapped structures in v2; 12 added since canonical v1. Binding/design context, not cellular occupancy.
Chemistry	Biochemical edge: SUPPORTED. Defined preparations and assay contexts only.
Cellular pharmacology	HLA-phenotype edge: SUPPORTED. Reporter/peptidome changes are not direct engagement.
Target engagement	Engagement edge: NOT_ASSESSED. Historical compound-specific attribution unresolved.
Disease translation	Disease-model edge: SUPPORTED; clinical edge: NOT_ASSESSED. No established human axSpA efficacy.

CURRENT DECISION

DECISION STABLE

Expanded evidence coverage did not change the critical uncertainty, next experiment or strategy identifiers. This is not a claim that all evidence or uncertainty states are unchanged.

CRITICAL UNCERTAINTY

Do the phenotype-producing compounds directly engage ERAP1 in cells at those exposures?

NEXT DISCRIMINATING EXPERIMENT

Matched-genotype chemical-genetic experiment with an engagement readout

Engagement at phenotype-active exposure, coupled to a matched null/rescue comparison, can strengthen or weaken compound-specific attribution. The proposal is not performed or investigator-approved; feasibility remains unknown.

Supporting evidence and constraints

Supporting continued investigation

ERAP1/HLA genetic associations motivate peptide-handling biology, not a prescribed intervention. [1,2]

Peptide trimming and HLA-B27 repertoire effects provide mechanistic context; cellular responses are system-specific. [3-6]

Allotype/substrate dependence is experimentally characterized and material to interpretation. [8,9]

Ligand-bound experimental regulatory-site structures and measured chemistry support design/tractability under recorded construct and assay contexts. [7,10-12]

BRADSHAW establishes defined Hap2/YTAFTIPSI biochemical activity and HeLa antigen-presentation pharmacology. Tinworth adds separate programme evidence; neither transfers occupancy to Maben compounds. [10-12]

Limiting the decision

Historical phenotype-producing Maben compounds lack demonstrated compound-specific direct cellular engagement in the curated decision. Off-target and indirect explanations remain viable. [7]

Biochemical inhibition, cellular antigen presentation, engagement and disease modification are different evidence classes.

Unreported context and allotype/substrate dependence restrict pooling and transfer. Chen and Tran report different free-heavy-chain directions in different systems. [5,6]

No established human axSpA therapeutic efficacy follows from these preclinical data. Independent review remains pending; no predictive model was trained.

Comparability is inference-scoped

Comparison	Frozen category
BRADSHAW internal biochemical	COMPARABLE
BRADSHAW biochemical vs cellular	NOT_COMPARABLE
BRADSHAW vs Tinworth biochemical	PARTIALLY_COMPARABLE
BRADSHAW vs historical Maben	NOT_COMPARABLE
BRADSHAW allotype/substrate panel	NOT_COMPARABLE

Chemical tractability is better established than therapeutic causality in this frozen corpus.

COMPARABLE is confined to source-specific Hap2/YTAFTIPSI SAR with exclusions; PARTIALLY_COMPARABLE establishes a method family, not quantitative interchangeability, selectivity ratios or pooling.

Why this experiment next?

Observed activity + cellular phenotype → competing explanations → unresolved engagement → matched engagement/null/rescue experiment → prospective outcome → decision consequence

Explanation	Frozen status / scientific alternative
Context-dependent effect	VIABLE: The effect depends on HLA-B27 subtype or ERAP1 allotype context rather than being a general consequence of ERAP1 modulation.
Indirect pathway	VIABLE: ERAP1 is perturbed, but the HLA-B27-related phenotype arises through an indirect pathway rather than proximal catalytic modulation.
Off-target activity	VIABLE: The phenotype arises primarily from compound activity at targets other than ERAP1 (for example ERAP2 or unrelated proteins).
On-target ERAP1 modulation	PARTIALLY_SUPPORTED: The HLA-B27-related cellular phenotype produced by the compounds arises from direct modulation of ERAP1 in those cells.

Proposed design - not an executed result

HLA-B27-positive cell system with wild-type, ERAP1-depleted or null, and ERAP1-rescued arms in a matched genetic background. Compound at phenotype-active exposures versus vehicle and an inactive analogue in each arm; ERAP1 genetic depletion and re-expression as the genetic comparison.

Primary endpoint: Compound-dependent ERAP1 engagement at the exposures that produce the phenotype, together with phenotype attenuation in ERAP1-null cells. **Context:** report/match ERAP1 allotype; choose/report HLA-B27 subtype. Measure viability separately and use the same phenotype-active exposures.

Controls and feasibility: vehicle; matched ERAP1-null/depleted cells; rescue; inactive analogue only if established (none identified). No qualifying positive-control ligand is established in the curated corpus. Compound identity, engagement readout and null/rescue validation remain prerequisites; cost/time and local feasibility are unknown.

Outcome → decision (all prospective AI suggestions)

ID	Observation	Decision consequence
s1	Engagement + attenuation in null + rescue	Strengthen this compound/context strategy; indirect downstream action and broader context are not excluded.
s2	Engagement + phenotype unchanged in null	Weaken ERAP1 attribution; alternative chemotypes require their own evidence.
s3	No signal in validated assay + attenuation in null	Change mechanistic model toward an indirect route; re-justify strategy.
s4	Engagement/dependency vary by background	Investigate matched allotype/HLA context before strategy choice.
s5	Invalid null/rescue or toxicity confounding	No mechanistic inference; validate controls/model and repeat.

A single system does not establish patient relevance. Null/rescue is not partial pharmacological inhibition; selectivity against other proteins remains unresolved. No scenario has occurred in this reassessment.

2026 reassessment: change without closure

What changed

The causal diff records expanded structural, chemical and cellular evidence coverage. Twelve source-mapped structures were added since canonical v1; these provide regulatory-site design context, not occupancy. [7,10-12]

Legitimate access to the BRADSHAW main article resolved the assay-identity blocker. Page 17/reference 22 explicitly links to Liddle 2020: purified Hap2, YTAFTIPSI cleavage and RapidFire MS. This chain supports constrained interpretation, not inferred equivalence based on programme similarity. [10,12]

Conservative BRADSHAW subsets retain 38 biochemical compounds/24 Murcko scaffolds and 31 cellular compounds/19 scaffolds. Scientific eligibility is MODEL_ELIGIBLE_WITH_CONDITIONS; no model was trained. Tinworth remains separate SAR-only evidence. [10,11]

What did not change

Historical Maben compound-specific engagement remains unresolved. Biochemical potency does not establish cellular engagement; a cellular phenotype does not establish a therapeutic mechanism. Later programme evidence cannot supply occupancy for earlier compounds.

Critical uncertainty and the recommended chemical-genetic engagement experiment remain unchanged. Strategy identifiers are stable. No protective human peptide state, optimal modulation magnitude or human axSpA efficacy is established.

Material restrictions retained

Unknown temperature, preincubation, dose range and tested form restrict exact replication and cross-programme comparison. HeLa ERAP1 allotype, ERAP2, endogenous HLA and control-inhibitor identity remain unconfirmed.

Compound 28 retains the main/SI discrepancy (cellular pIC50 5.3 versus 5.4). Censored occasions for 21/28/33/44, singleton observations and missing N are preserved and excluded where required for fitting. No bound becomes an exact value.

Epistemic key

Source assertion: what a publication reports. **AXIS observation:** recorded extraction/calculation. **AXIS inference:** scoped interpretation. **AI suggestion:** competing explanations/experiment/scenarios. **Researcher hypothesis:** proposed mechanism. **Experimental result:** an observed performed experiment, not a prospective scenario. These kinds are not interchangeable; pending reviews remain pending.

The disease-model edge is SUPPORTED in v2 while clinical translation remains NOT_ASSESSED. Animal/cellular evidence is not human axSpA efficacy; the stable decision concerns the still-missing compound-specific engagement link.

Independent critique and provenance

- 1. Is the ERAP1/HLA-B27/spondyloarthritis biology represented accurately?
- 2. Is any material evidence overstated?
- 3. Is important evidence missing within the stated scope and cutoff?
- 4. Are assay contexts and comparability restrictions appropriate?
- 5. Is allotype/substrate dependence represented appropriately?
- 6. Are biochemical activity, phenotype and direct engagement sufficiently distinct?
- 7. Is historical Maben evidence correctly separated from later programme tractability?
- 8. Is direct cellular engagement the most important unresolved decision uncertainty?
- 9. Does the chemical-genetic experiment discriminate the major explanations?
- 10. Is DECISION STABLE a scientifically defensible interpretation?

What is the strongest scientific reason this decision could be wrong?

Field	Response (repeat for each issue)
Severity	CRITICAL / MAJOR / MINOR / NO_ISSUE
Claim/artifact	Provide exact locator
Evidence/source	Provide exact locator
Rationale	Explain the scientific concern
Recommended correction	Describe the remedy
Changes decision?	YES / NO / UNCERTAIN; explain

Review snapshot

Cutoff 2026-10-05 | Decision: DECISION STABLE
Scientific/runtime: d11128538a2723737eced46be73d40cef56f43db
Scientific reference: 84ecc56e74747bdee12ecdf15b9a80bc760798f4
v1: decision-state:1:cebbae3880aeec90 | v2: decision-state:2:212f995ecc55c989
Critical: uncertainty:target_engagement
Next: decision:exp:chemical-genetic-engagement

Every material AXIS conclusion is intended to be traceable from conclusion → claim → evidence → experiment/source. This is an intended audit chain, not certification of completeness or factual accuracy.

- Current scientific state - immutable d111285
- BRADSHAW reassessment - immutable d111285
- DecisionState v2 - immutable d111285
- Causal diff - immutable d111285
- Assay comparability - immutable d111285
- Evidence manifest - immutable d111285
- Independent-review handoff - immutable 54ea4ef

Key primary sources

[1] Interaction between ERAP1 and HLA-B27 in ankylosing spondylitis implicates peptide handling in the mechanism for HLA-B27 in disease susceptibility. (2011).

DOI: [10.1038/ng.873](https://doi.org/10.1038/ng.873)

Frozen discovery provenance; access is claim-specific.

[2] Major histocompatibility complex associations of ankylosing spondylitis are complex and involve further epistasis with ERAP1. (2015).

DOI: [10.1038/ncomms8146](https://doi.org/10.1038/ncomms8146)

Frozen discovery provenance; access is claim-specific.

[3] The ER aminopeptidase, ERAP1, trims precursors to lengths of MHC class I peptides by a "molecular ruler" mechanism. (2005).

DOI: [10.1073/pnas.0500721102](https://doi.org/10.1073/pnas.0500721102)

Frozen discovery provenance; access is claim-specific.

[4] Critical role of endoplasmic reticulum aminopeptidase 1 in determining the length and sequence of peptides bound and presented by HLA-B27. (2014).

DOI: [10.1002/art.38249](https://doi.org/10.1002/art.38249)

Frozen discovery provenance; access is claim-specific.

[5] Silencing or inhibition of endoplasmic reticulum aminopeptidase 1 (ERAP1) suppresses free heavy chain expression and Th17 responses in ankylosing spondylitis. (2016).

DOI: [10.1136/annrheumdis-2014-206996](https://doi.org/10.1136/annrheumdis-2014-206996)

Frozen discovery provenance; access is claim-specific.

[6] ERAP1 reduces accumulation of aberrant and disulfide-linked forms of HLA-B27 on the cell surface. (2016).

DOI: [10.1016/j.molimm.2016.04.002](https://doi.org/10.1016/j.molimm.2016.04.002)

Frozen discovery provenance; access is claim-specific.

[7] Discovery of Selective Inhibitors of Endoplasmic Reticulum Aminopeptidase 1. (2020).

DOI: [10.1021/acs.jmedchem.9b00293](https://doi.org/10.1021/acs.jmedchem.9b00293)

Historical biochemical/cellular compounds; not direct cellular occupancy.

[8] Common allotypes of ER aminopeptidase 1 have substrate-dependent and highly variable enzymatic properties. (2021).

DOI: [10.1016/j.jbc.2021.100443](https://doi.org/10.1016/j.jbc.2021.100443)

Frozen allotype/substrate or cellular immunopeptidome context.

[9] Distinct modulation of cellular immunopeptidome by the allosteric regulatory site of ER aminopeptidase 1. (2023).

DOI: [10.1002/eji.202350449](https://doi.org/10.1002/eji.202350449)

Frozen allotype/substrate or cellular immunopeptidome context.

[10] Automated Molecular Design in BRADSHAW, Applied to the Optimization of ERAP1 Inhibitors (2026).

DOI: [10.1021/acs.jmedchem.5c03071](https://doi.org/10.1021/acs.jmedchem.5c03071)

Law et al.; main p4-5, p17/ref22; SI S60/Table S2. Institutional access; PDF not redistributed.

[11] Discovery of an Orally Available Potent ER Aminopeptidase 1 (ERAP1) Inhibitor That Enhances Antitumor Responses and Limits Inflammatory Autoimmunity In Vivo. (2026).

DOI: [10.1021/acs.jmedchem.6c00029](https://doi.org/10.1021/acs.jmedchem.6c00029)

Tinworth programme. Methods rely on accessible preprint 10.1101/2025.11.17.686761 v1/SI; final main inaccessible in frozen scope. One publication chain, not independent replication.

[12] Liddle et al. - ERAP1 biochemical assay-method source (2020).

DOI: [10.1021/acs.jmedchem.9b02123](https://doi.org/10.1021/acs.jmedchem.9b02123)

Bibliographic locator retained without inventing an unrecorded title; main p8(H), linked explicitly by BRADSHAW p17/ref22.

Access and scope

References are selected from the frozen corpus, not a new systematic search. Metadata/abstract access is not full-text adjudication. The later BRADSHAW v3 access record supersedes its older insufficient-access label; Tinworth final-main restrictions remain. Licensed primary PDFs are not included. Missing engagement in this corpus is not a claim that no engagement evidence exists anywhere.

Please return structured criticism of the decision and its provenance. Independent scientific review remains pending.